

Chapter 23

Towards a typology of Bantu complementizers

John Gluckman
University of Kansas

This paper provides a preliminary overview of the lexical sources and functions of complementizers used for finite, non-interrogative, syntactically selected embedded clauses in Narrow Bantu languages. It identifies five distinct sources for complementizers in Bantu languages: ‘say,’ ‘be,’ demonstratives, manner deictics, and pronouns. The paper further discusses languages which have more than one such complementizer, and investigates three factors that correlate with complementizer choice. Based on these initial findings, the paper concludes that there is a general dichotomy between complementizers, which is framed in terms of “perspectival” vs. “situational” anchoring.

1 Introduction

In this paper, I provide a preliminary typology of *complementizers* in the Narrow Bantu languages.¹ This typology covers three aspects. First, I address the lexical source of complementizers across Bantu languages, building on typological work in e.g., [Cristofaro \(2003\)](#), [Kuteva et al. \(2019\)](#). Second, Bantu complementizers are widely documented as having various “effects” on the utterance. I illustrate some of those effects, noting the range of syntactic, semantic, and morphological

¹The classification “Narrow Bantu” applies to roughly 600 Bantoid languages, ranging from West to East Africa and into Southern Africa. Narrow Bantu specifically excludes the Grassfields Bantu languages of Western Africa. The classification of individual Bantu languages follows the Guthrie code system (updated in [Hammarström 2019](#)), which consists of one or two letters indicating sub-family and number indicating the specific language in that sub-family. In this paper, Guthrie codes are given following each language.

variation. I finally conclude by looking briefly at some correlations between the lexical source of a complementizer and the effect it can have, observing that there is a general split among the Bantu complementizers.

Based on these preliminary findings, I suggest an informal classification in terms of *anchoring*, that is, how the embedded clause is integrated into the superordinate material. “Perspectival” complementizers anchor the embedded clause to a relevant *individual*; “situational” complementizers anchor the embedded clause to a relevant *situation* (or event). Ultimately, I suggest that all Bantu complementizers fall into one of these two camps – at least historically.

Before beginning, a note on terminology is in order. I use the term *complementizer* atheoretically to cover a number of potentially disparate categories, including clausal conjunction and (clausal) subordinator. I also assume that there is no formal distinction between a complementizer and a quotative marker (Güldemann 2002).

Finally, I am confining my investigation to finite noninterrogative clausal complementation. Thus, I put aside nonfinite complementation and interrogative complementation, as well as other clause-types, like relative clauses, causal clauses, temporal clauses, conditional clauses, etc. This is purely to constrain the scope of the study.²

2 Sources of complementizers

Crosslinguistically, a number of distinct sources for complementizers are recognized (Hopper & Traugott 1993, Güldemann 2008, Kuteva et al. 2019). Chappell (2008) summarizes the known sources in (1), with the bolding to indicate sources that Bantu languages instantiate.

- (1) a. nouns such as ‘thing’, ‘fact’ or ‘place’
- b. **demonstrative**, interrogative and **relative pronouns**
- c. dative, allative and locative case markers or prepositions
- d. **SAY verbs**

²The data presented here draws on over 100 languages (not all of them cited here for space) and is collected mainly from existing descriptive and theoretical sources on Bantu languages. Additional data comes from the authors’ fieldnotes. I emphasize the preliminary nature of this study; more data are needed to paint an accurate picture. An attempt was made to include data from all of the Guthrie Zones (the classificatory system for Bantu languages; I will rely on Hammarström’s (2019) updated Guthrie codes). However, Eastern Bantu is over-represented in the language sample due to the prevalence of material on those languages. When a citation is not present, the data was collected through fieldwork by myself.

- e. similitive verbs meaning ‘resemble’ or ‘be like’
- f. **similitive manner adverbials and deictics**

adapted from Chappell (2008: 3)

As expected, in these data, we find considerable variation between languages (and even within languages) in the form of the complementizer and its diachronic or synchronic source. Bantu languages robustly instantiate at least three of these tendencies: demonstratives (including pronominal complementizers), *say*-verbs, and manner deictics. Bantu languages additionally employ a strategy not described in (1) above: *be*-complementizers, i.e., complementizers derived from or related to the copula. As these are distinct from similitive verbs and particles (the latter also attested to a lesser extent in these data), I treat *be*-complementizers as a distinct lexical source. I note that my goal below is not to provide a diachronic trajectory for these transitions; I will not explain *why* certain lexical categories are recruited as complementizers. I believe that this motivation has been fleshed out for most categories in Kuteva et al. (2019) and related work – save *be*-complementizers, which I leave for future work.

2.1 Say-complementizers

The appearance of *say*-complementizers is widespread across the world’s languages (cf. Hopper & Traugott 1993: 14; Kuteva et al. 2019: 375). Accordingly, a large number of Bantu languages employ a form of the verb meaning, or diachronically related to, ‘say.’ In (2a), Mwani’s (G403) *kamba* is derived from proto-Bantu **(g)amb* ‘word, speak’ (cf. Meeussen 1967: 94). In (2b), Bena’s (G63) *uhutigila* functions synchronically as a *say*-verb, as does Kisi’s (G67) *kujobha*, in (2c). (Note that, despite the different lexicalizations, these are all Zone G languages.)³

- (2) a. Mwani (G403) (LIDEMO 2010: 55)
 Amadi akwijiwa kamba Ali kawa nao nzuruku.
 Amadi knows COMP Ali be with money
 ‘Amadi knows that Ali had money.’

³Here and throughout, examples are provided with the glossing as given in the source material. See Section 5 for a list of abbreviations.

- b. Bena (G63) (Morrison 2011: 417)
a-va-anu va-i-pulih-a uhutigíla u-mu-yeesu
AUG.2-CL2-person CL2-PRES-hear-FV COMP AUG.1-CL1-our.friend
a-taagih-ile
CL1-die-FV
‘People hear that our friend died.’
- c. Kisi (G67) (Nicolle et al. 2018: 13)
Lingi.na.lingi. bha-ka-n’-jobh-el-a kujobha, “Bit-agh-e
after.a.while 3PL-CONS-3SG-say-APPL-FV QUOT go-IPFV-SBV
ka-kot-agh-e ku=bhulongolo
ITIVE-ask-IPFV-SBV 17.LOC=ahead
‘After a while they said to him, “Go and ask up ahead...”’

The examples above illustrate that multiple languages have *independently* developed a *say*-complementizer from distinct sources. However, extending across nearly all subfamilies are derivatives of proto-Bantu **-tì*, which in many languages has evolved into an element meaning ‘say’ as well as a complementizer (Meeussen 1967, Güldemann 2002). Note that for many languages it is used primarily to introduce direct speech, and thus has a quotative function (Güldemann 2008). The examples in (3) demonstrate an infinitival, thus verbal, form of **-tì*.⁴

- (3) a. Nyakyusa (M31) (Persohn 2017: 314)
ɔ-meenye ɔkɔtɪ Asia a-ka-kɔ-gan-a?
2SG-know.PFV COMP A. 1-NEG-2SG-love-FV
‘Do you know that Asia doesn’t love you?’
- b. Zulu (S42) (Halpert 2016: 1)
ku-bonakala ukuthi uZinhle u-zo-xova ujeqe
17s-seem that AUG.1Zinhle 1s-FUT-make AUG.1steamed.bread
‘It seems that Zinhle will make steamed bread.’
- c. Fwe (K402) (Gunnink 2018: 432)
mbo-á-bo_Hn-é kutí Ø-ci-pepa bu-ryó
NEAR.FUT-SM₁-see-PFV.SBJV COMP COP-NP₇-paper NP₁₄-only
ci-bá-mu-dara
PP₇-NP₂-NP₁-old.man
‘She will see that it is just a paper of her husband.’

⁴Gunnink (2018) explains that in Fwe, the form *kutí* and the variant *kûtêyè* are “contractions of the verb *kùtá* ‘to say’, with the complementizer *iyé*.”

In the data, derivatives of **-ti* are prevalent in Eastern and Southern families. Though often correlating with a verb of speech in modern languages, [Güldemann \(2002, 2008\)](#), following the reconstruction in [Guthrie \(1970b: 105\)](#), argues that the original function of **-ti* was that of a manner deictic, a demonstrative element that is anaphoric (or cataphoric) to an expression of manner or kind. Over time, in some languages, the manner deixis meaning transitioned to a speech verb (often defective). The manner deictic use of **-ti* is still found in a number of Bantu languages; I return to this in Section 2.3.2.

Still, it is undeniable that some languages have derivatives of **-ti* which are verbal.⁵ This is clear from the fact that many *ti*-based complementizers may still bear vestiges of verbal inflectional marking, like tense, aspect, and voice morphology ([Persohn 2017](#), [Letsholo & Safir 2019](#), [Dembetembe 1976](#)).

However, it is sometimes unclear how much of this inflectional material is completely fossilized as part of the complementizer. For instance, should Xhosa's (S41) invariant complementizer *sengathi* be analyzed synchronically as consisting of *se-* 'aspect', *nga-* 'possibility modal', and *-thi* 'say' (cf. [Halpert 2019: 33](#))? Moreover, it is also clear that some *say*-complementizers have nominal properties. This is the core finding of [Pietraszko's \(2019\)](#) work on the Ndebele (S44) complementizer *ukuthi* (again derived from **-ti*), which is argued to be housed inside of a nominal layer.

Finally, I note that [Devos & Bostoen \(2012\)](#) discuss a variant of the *say*-complementizer strategy: some Bantu languages employ a complementizer that is related to (again, either synchronically or diachronically) a word meaning 'do'.⁶ This is shown for Shangaci (P312) in (4) with the verb *-ira*, translated variously as 'do, say, (be) like.' As reported in [Devos & Bostoen \(2012\)](#), the same polysemy is found in Lunda (L52), Nkore-Kiga (JE13/14), Rundi (JE62), Rwanda (JE61), and Mongo (C61).

(4) Shangaci (P312) ([Devos & Bostoen 2012: 99](#))

- a. ki-ao-ir-i <ni-law-e o-mu-ti>
 SC_{1SG}-PST-QV-PFV <SC_{1pl}-gO-SBJV NP₁₇-NP₃-town>
 'I said: 'Let's go home.'

⁵A reviewer notes that there are multiple paths to being a verb for **-ti*. It could develop verbal properties and then become a complementizer, or alternatively, it could be co-opted as a quotative and then develop verbal properties subsequently.

⁶The data reported here draws heavily on the discussion in [Devos & Bostoen \(2012\)](#). Any errors in reporting and interpreting their data are my own.

- b. ir-an-i tono mu-mu-thuul-e ontu
 QV-IMP-PLA thus SC_{2p1}-OC₁-take-SBJV PP₁.DEM_i
 ‘Do it as follows: Take this one...’
- c. mu-khira o-awe o-sal-a o-ir-a khong’ongoo
 PP₃-tail PP₃-POSS₁ PP₃-remain-PFV PP₃-QV-PSIT IDEO
 ‘His tail remains *like* a stump.’

It is likely that there is a single proto-Bantu source for *do*-complementizers in Bantu: *-gid- (Bastin et al. 2002). As noted in Devos & Bostoen (2012), given the polysemy of *-gid- across Bantu, it may ultimately be possible to list *do*-complementizers as a sub-type of *say*-complementizer.⁷

2.2 *Be*-complementizers

In this study, I found a number of Bantu languages which introduce embedded clauses using a form of the copula.⁸ In the examples in (5), all the complementizers are *synchronically* multi-functional as infinitival forms of a verb meaning ‘be.’

- (5) a. Haavu (JD52)
 Mugisha a-yubah-ire okuba o-lu-juki lu-a-mu-lume.
 Mugisha 1SM-be.afraid.PFV COMP 11AUG-11NC-bee 11SM-FUT-1OM-sting
 ‘Mugisha is afraid that the bee will sting her.’
- b. Ikoma (JE45) (Roth 2018: 94)
 nihó Núhu a-ka-meny-a kuḃá amánche
 so Noah 3SG-NARR-come.to.know-FV that waters
 ya-ayá-tiḃok-a mose
 6-NUCL-decrease-FV on.land
 ‘So Noah knew that the waters had subsided from the earth.’

⁷An interesting though perhaps exceptional case is Mwimbi-Muthambi (E531), which uses *taka*, plausibly derived from a verb meaning ‘want’ as the complementizer (Kaburo 2022). If this is truly the etymology of the complementizer, then it may also be grouped together with the other “action”-oriented complementizers like *say* and *do*.

⁸I refer the reader to the discussion in Gibson et al. (2018), Finholt (2024) concerning the varieties of copular predication in Bantu. This is a complex topic. As it applies to complementizers, it is clear from the data collected that there are a limited number of *be*-complementizers, related to only a subset of possible copulas across Bantu.

- c. Xhosa (S41) (du Plessis 1989: 44)
 ndafumana ukuba mandifunde isiXhosa
 I.found that I.must.study Xhosa
 ‘I found that I must study isiXhosa.’
- d. Mbundu (H21a) (Johnson 1931: 77-78)
 mwane wambe kuma “eme maza ngendele mu-nyanga”
 he said that I AUX hunt yesterday
 ‘He said “I went hunting yesterday.”’

The majority of *be*-complementizers in our sample appear to be derived from proto-Bantu **-bá* ‘be, become’ (Guthrie 1970a: 17; Meeussen 1967: 86), retaining a bilabial place feature (/b, β, m, w/) in all subsequent forms.⁹ Still, there are some languages, particularly those in Zone E, which appear to have independently developed complementizers from other forms of the copula. Digo (E73) and Giryama (E72a) both use *kukala* ‘be.’ This verb is likely derived from proto-Bantu **-kàd* ‘dwell, be’ (Guthrie 1970a: 258) or **-yìkad* ‘dwell, be’ (Guthrie 1970b: 179), with the d/l alternation, already attested in proto-Bantu. Likewise, I believe that *kana*, found in a few disparate languages like Kamba (E55) has a similar origin, with **d>n*.¹⁰

- (6) a. Digo (E73) (Nicolle 2014: 55)
 Ndipho atu a-chi-many-a kukala iye ndi=ye
 then 2-people 3PL-CONS-know-FV COMP 3SG COP=1.REF
 a-ri-ye-hend-a mambo higo.
 3SG-PST-1.REL-do-FV 6.things 6.DEM.NP
 ‘then people knew that it was her who did those things.’
- b. Giryama (E72a) (Lax 1996: 273)
 yuyu ŋmanamuče were walagiza kukala a-so-zìkp-e
 that woman AUX instructed COMP 1SM-NEG-bury.PASS-SBJV
 ‘The woman had instructed that she should not be buried.’
- c. Kamba (E55) (Myers 1975: 186)
 aisye kana nũkũka ũmũnhi
 he.say.PAST that he.come.PRES tomorrow
 ‘He said that he’s coming tomorrow.’

⁹Johnson (1931: 78) observes that *kuma* is an “impersonal expression,” generally used for “there/it is...” The source of the nasality here is not clearly **b* however.

¹⁰The **d>n* trajectory is not common, but it is not unheard of either. Guthrie (1970a) lists reflexes of **-yìkad*/**-kàd* as **-jkan* and **-jan* in East Holoholo (D28b) and Bembe (D54) respectively.

Beyond the copular elements above, I found a few Bantu languages that use the morpheme *n(i)* as complementizer. In Bembe (D54) in (7a), *ni* appears with an invariant class 8 form of the relative pronoun *-bo* (and thus also falls under the pronominal category of complementizer, discussed in Section 2.3.3). In Luba-Kasai (L31a), I believe that *ne* is the cognate form in (7b).

- (7) a. Bembe (D54)
Aachi a-le-ngany-a nibo Mswakeecha a-le mu-lwaala.
Aachi 1SM-PRES-think-FV COMP Mswakeecha 1SM-COP 1AGR-sick
'Aachi thinks that Mswakeecha is sick.'
- b. Luba-Kasai (L31a)
Kalombo mu-sw-e ne Mujinga a-y-e ku Tshinsansa.
Kalombo 1SM-want-FV COMP Mujinga 1SM-go-SBJV to Kinshasa
'Kalombo wants Mujinga to go to Kinshasa.'

Across Bantu, "[as] proclitic or prefix, [ni] has a number of functions easily relatable to the copula" (Nurse 2008: 53). I thus choose to list examples like (7) as *be*-complementizers. However, it is also observed that *ni* is strongly associated with *focus* (Nurse 2008, Güldemann 2013). Therefore, we may ultimately want to list *ni*-based complementizers as **focus complementizers**. Still, there is a tight connection between copulas and focus (Kuteva et al. 2019: 125), whereby focus markers are often grammaticalized from copulas. Therefore, it is not surprising that we see a link between focus and complementizers; however, it is unclear whether that link is always through the shared connection with copulas.

2.3 Deictic complementizers

Bantu languages contain a diverse range of complementizers related to deixis. I divide up the class of deictic complementizers into three distinct subclasses: demonstrative, manner, and pronominal deixis.

2.3.1 Demonstrative complementizers

Our data reveals that a number of Bantu languages employ a complementizer that elsewhere has (or had) a demonstrative function. All languages in Zone JD have as a complementizer a form of the class 15 form of the proximal demonstrative, realized as *uku* 'this₁₅' in Rwanda (J61).

- (8) a. Shi (JD53) Aron Finholt (p.c.)
Mugisho a-lá:-waz-a ku Murhula a-li Bujumbura.
Mugisha 1SM-PRES-think-FV COMP Murhulla 1SM-COP Bujumbura
‘Mugisho thinks that Murhula is in Bujumbura.’
- b. Rundi (JD62) (Sabimana 1986: 200)
y-a-vuz-e ko Maria y-a-ri u-mu-nyeshuri
1-RPAST-say-ASP that Mary 1-FPAST-be A-1-student
‘He said that Mary was a student.’

Lega’s (D25) complementizer is homophonous with the proximal demonstrative *-no* with class 14 marking, *bo-*.

- (9) Lega (D25) (Botne 1995: 214)
ámábúnde bóno ékwendá ko Zaíle
3s.1s.tell.PST that 3s.PR-go to Zaire
‘S/he told me that s/he is going to Zaire.’

In our data, deictic complementizers always show invariant noun class inflection, though the form is not predictable for any one language as far as I can tell. However, based on the available data, some generalizations can be made. Specifically, demonstrative complementizers broadly reflect noun classes that are otherwise used for abstract concepts and/or mass terms. For instance, Class 15 in JD60 is used elsewhere exclusively for infinitival forms of the verb. Class 14 in Lega is used for abstract nouns (and also gelatinous substances and body parts; Botne 2003). In Mbugwe (F34), the complementizer *kóko* is recorded as being identical to the class 17 locative form of the proximal demonstrative (Mous 2004).

2.3.2 Manner deictics

Included among the deictic complementizers are manner deictics. As observed originally in Guthrie (1970b: 105) (and argued for explicitly in Güldemann 2002, 2008), this is probably a relevant derivative function of proto-Bantu **-ti*. Though a speech verb in some Bantu languages (Section 2.1), in fact Guthrie traces the original meaning to ‘that, namely’, which later in some languages is “best expressed in English as ‘saying’” (Guthrie 1970b: 105). In many Bantu languages, though, the manner deictic use persists. For instance, in Nyala East it is clear that *-chi*,¹¹ has the distribution of a manner deictic outside of embedding contexts.

¹¹This is, again, historically **-ti*. In much of JE30, stops (af)fricated over time (Guthrie 1970a)

- (10) a. Nyala East (JE32f)
o-mw-aana a-chi
1AUG-1NC-child 1SM-DEM
'a child like this'
- b. Nyala East (JE32f) (Gluckman 2023)
Masika a-paar-a a-chi Wekesa ka-chi-a Nairobi
Masika 1SM-think-FV 1AGR-COMP Wekesa 1SM-go-FV Nairobi
'Masika thinks that Wekesa went to Nairobi.'

It is possible that manner deictics other than *-tì have independently developed into complementizers. In Eton, *nâ* introduces selected embedded clauses as well as anaphorically describes a manner.

- (11) a. Eton (A71) (Van de Velde 2008: 351)
à-Lté L-kâd H b-òd nâ H-bâ-zù-L
I-PR INF-tell LT 2-person CMP SB-II-come-SB
'He tells the men to come.'
- b. Eton (A71) (Van de Velde 2008: 170)
mâ-Lté kôm nâ
1SG-PR INF-do thus
'I do it this way.'

2.3.3 Pronominal complementizers

A number of Bantu languages have developed complementizers from ostensibly pronominal sources, a pattern that is well-attested outside of Bantu languages (Diessel & Breunese 2020). This is explicitly argued by Botne (1995) to be true of the complementizers *ngo* and *mbo*, found throughout Bantu languages.

- (12) a. Rwanda (JD61) (Rugege 1984: 38)
Mary ya-inubwa ngo abana ba-tah-e kare.
Mary 1SM-resent COMP 2NC.children 2NC-get.home-SBJV early
'Mary resents it that the children get home early.'
- b. Nyambo (JE21) (Rugemalira 2005: 96)
akajira ngu muri abasúma
she.imagine.PAST COMP you.were thieves
'She imagined that you were thieves.'

- c. Ruuli (JE103) (Sørensen & Witzlack-Makarevich 2020: 94)
 naye abandi ba-kob-a mbu o-Bunyala bu-a-ikang-a
 but AUG.2.other 2S-say-FV COMP AUG-Bunyala(14) 14S-PST-reach-FV
 o-ku Nammanve
 AUG-17.LOC Nammanve(9)
 ‘But others says that the Bunyala reached Nammanve.’
- d. Mongo (C61) (De Rop 1958: 126)
 básanga mbo basúwa bāosil’ókenda
 2SM.claim that boats 2SM.left
 ‘They claim that the boats have left.’¹²

Botne (1995) argues that these forms are historically related to (emphatic) pronouns. Note that, since they derive from pronouns, pronominal complementizers are often “agreeing,” in that they co-vary with the subject of the embedding verb. Compare the “discourse markers,” which can be used to introduced any kind of embedded clause (and are preceded by the “particle” *a*), with the pronominal forms in Table 1. An example of the discourse particle use is shown in (13).

Table 1: Ntomba (C35a) (Botne 1995: 209)

	Direct discourse markers		Personal pronouns	
	Singular	Plural	Singular	Plural
1	<i>a mi</i>	<i>a nso</i>	<i>mi</i>	<i>so</i>
2	<i>a we</i>	<i>a nyo</i>	<i>we</i>	<i>nyo</i>
3	<i>a nde</i>	<i>a mbo</i>	<i>nde</i>	<i>bo</i>

- (13) Ntomba (C35a)] (Gilliard 1928: 69), cited in (Botne 1995: 208)
 bayoye batepela, a mbo: ntaba inkuma isolota
 3P-come-PST 3SP-say-FV PRT 3P 10-goats 10-all 10-PRF-run.off
 ‘They came and said: all the goats have run off.’

Not all Bantu pronominal complementizers are related to *ngo* or *mbu*. In particular, Kawasha (2006, 2007) reports that complementizers in Chokwe (K11), Luchazi (K13), Lunda (L52) and Luvale (52) are directly related to the possessive pronouns in those languages.

¹²In original, *Ze beweren, dat de boten vertrokken zijn*.

2.4 Other categories

The above categories (*say*-complementizers, *be*-complementizers, deictics) were found fairly robustly throughout these data. I report below on two additional strategies that I feel should be mentioned, but did not appear as frequently in our language sample.

First, there are *similatives*, words meaning “like” or “as” Kuteva et al. (2019). In this study, we find derivatives of **-ngà* ‘as, like’ (Guthrie 1970b: 243) serving as complementizers, like Ruuli (JE103), shown in (14).

- (14) Ruuli (JE103) (Sørensen & Witzlack-Makarevich 2020: 94)
a-baana a-iz-a ku-bon-a nga ba-ku-ikiriz-a
AUG-child(2) 2SG.S-AUX-FV INF-see-FV COMP 2S-2SG.O-believe-FV
‘You will see the children believe you.’

Second, I feel obligated to mention that not all finite embedded clauses require an overt complementizer. In many cases, the complementizer can be omitted, with no discernable meaning difference, like in Kikuyu (E51), shown in (15).

- (15) Kikuyu (E51) (Englebretson 2015: 150)
mũ-timia a-kũ-ĩtik-ĩt-i-e (atĩ) mũ-thuri
NC₁-woman SC₁-CR.PST-believe-PERF-TRNS-FV COMP NC₁-man
nĩ-a-Ø-iy-ir-e N-gũkũ
FOC-SC₁-CR.PST-steal-COMPL-FV NC₉-chicken
‘The woman believed (today) (that) the man stole the chicken.’

Very little documentary work addresses whether null complementizers are possible in a given language – though see Edelsten et al. (2022: section 3.10).¹³ Nonetheless, in at least some languages, all finite, declarative embedded clauses must be headed by an overt complementizer. For instance, as reported in Masatu (2015: 8) for Simbiti (JE431), “Both direct and indirect speech is marked with the complementizer *igha* ‘that.’ This complementizer appears with almost every occurrence of direct or indirect speech, and it seems to be extremely ungrammatical to omit it.”

¹³Edelsten et al. (2022) ultimately suggest that optionally expressed complementizers are more likely in East African Bantu languages. Thus, outside of East Africa, Bantu languages are more likely to have embedded clauses with obligatorily overt complementizers.

3 Functions of Bantu complementizers

We now turn to the functions of Bantu complementizers. This is a pertinent question because in our sample, many languages have multiple complementizers. Conservatively, 40 languages out of 103 have two or more complementizers used in finite non-interrogative embedding. What factors then affect the distribution of the complementizers within a language? And what factors influence when a speaker chooses one complementizer over another? I discuss here three factors that correlate with complementizer choice: commitment (“evidentiality”), information structure, and agreement. This is far from an exhaustive list. However, these three factors have been discussed in a number of sources, and so are amenable for making cross-linguistic comparison. Still, I emphasize that the documentary picture remains fairly weak when it comes to the functions of complementizers in Bantu languages, and so my findings here should be taken as preliminary. Nonetheless, based on the factors discussed below, I motivate a two-way distinction among Bantu complementizers. Across all three factors, *say*-complementizers, manner deictics, and pronominal complementizers pattern similarly, while *be*-complementizers and demonstratives pattern similarly.

3.1 Commitment (“evidentiality”)

Many sources that document Bantu language complementizers mention “evidentiality” (see in particular Botne 1995, 2020), with the caveat that the term “evidentiality” covers a lot of ground. Rather than evidentiality *per se*, Bantu complementizers more often reflect an individual’s – often the speaker’s – commitment to the embedded proposition (cf. Gluckman 2021, 2023). For example, in Kihara (2017), Kikuyu’s (E51) complementizer *atĩ* is argued to function as a dubitative marker (in some contexts). The speaker is “non-committal about the relayed information, rendering the information unreliable” (Kihara 2017: 114). In this way, *atĩ* is distinct from the “true” quotatives *atĩrĩ* and *atĩrĩrĩ*.

- (16) Kikuyu (E51) (Kihara 2017: 114)
 ndĩ-ra-igu-ir-e atĩ nĩ ma-ra-cok-ir-e ka-ao
 1SG-RCPST-hear-PFV-FV DUB AM 2-RCPST-return-PFV-FV 16-theirs
 ‘I heard that they returned to their home.’

In Givón & Kimenyi’s (1974) study of Rwanda’s (JD61) complementizer system, they note a contrast between *kó* and *ngo* (which are not in complementary distri-

bution). Similar to Kikuyu above, the pronominal complementizer *ngo* gives rise to a reading of speaker-doubt.

- (17) Rwanda (JD61) (Givón & Kimenyi 1974: 96)
 ya-m-bgiye ngo u-a-koraga cyaana
 he-PAST-me-tell ngo you-PAST-work-HAB hard
 ‘He told me that you worked hard (but I personally doubt it).’

This contrasts with the demonstrative complementizer *kó*, which, in some contexts, correlates with stronger speaker belief. For example, with a verb like *kubeeka* ‘to lie,’ with *kó* (as opposed to *ngo*) in (18), the speaker’s commitment to “the belief ‘John didn’t come’ is the **weakest**” (Givón & Kimenyi 1974: 102, emphasis in original).

- (18) Rwanda (JD61) (Givón & Kimenyi 1974: 102)
 ya-mu-beešye kó Yohani yaaže
 1SM-1OM-lie.PERF COMP John come.PERF
 ‘She lied to him that John came.’

Thus, with *kó*, the speaker is more committed to the truth of the embedded clause – even contradicting the lexical semantics of the embedding predicate.

Another example of stronger speaker commitment is observed in Kamba (E55), for which Myers (1975) reports that the *be*-complementizer *kana* (as opposed to no complementizer) in (19) indicates “the speaker believes that the promise will be kept” (Myers 1975: 190).

- (19) Kamba (E55) (Myers 1975: 190)
 aisye kana nūkūka ūmūn thi
 he.said that he.comes today
 ‘He said that he is coming today.’

A parallel distinction is reported in Finholt & Gluckman (2023). In Swahili (G42), the *be*-complementizer *kuwa* contrasts with the *say*-complementizer *kwamba* under verbs of epistemic possibility. When the speaker is relatively sure of the outcome, *kuwa* is better (20); otherwise *kwamba* is preferred (21).

- (20) *We’re watching Tanzania play in a football [soccer] match. There are five minutes left to play, and Tanzania is up by three.*
 i-na-onekan-a kuwa/#kwamba Tanzania i-ta-shind-a
 9SM-PRES-seem-FV COMP/COMP 9Tanzania 9SM-FUT-win-FV
 ‘It seems like Tanzania will win.’

- (21) *We're watching Tanzania play in a football [soccer] match. It's halftime, and Tanzanian is up one to nil.*

i-na-onekan-a kwamba/#kuwa Tanzania i-ta-shinda
 15SM-PRES-seem-FV COMP/COMP 9Tanzania 9SM-FUT-win
 'It seems like Tanzania will win.'

We therefore find both weak and strong commitment readings associated with complementizer choice. Interestingly, there are correlations between the lexical source of the complementizer and the associated reading. When we find a weak reading, it is always in the presence of a *say*-complementizer, manner deictic, or pronominal complementizer. Strong readings only ever arise in the presence of *be*-complementizers and demonstratives.¹⁴ Note that this does not entail that all complementizers have a “commitment” meaning component; the observation is simply that when such a meaning arises (in embedded clauses), the lexical source of complementizer is largely predictable.

3.2 Information structure

Complementizers often interact with the information structure of the embedded clause (cf. [Djäv 2019](#)). In Bantu languages, the clearest instance of such “embedded clause effects” is observed in the presence of *predicate focus*, i.e., a morpho-syntactic configuration in which the predicate (or something inside of the predicate) is in focus. This distinction is often seen in Eastern and Southern Bantu languages. In some of those languages, certain complementizers prohibit the predicate focus operator *ni-* found in many Bantu languages ([Güldemann 2003, 2013](#)) (cf. Section 2.2). In Subi (JE26),¹⁵ the predicate focus operator is banned in clauses introduced by *nkikwo*, but permitted in clauses introduced by *ngu*.¹⁶

¹⁴There is an interesting wrinkle here, however. *Be*-complementizers are often used to head embedded polar interrogatives (‘if/whether’), which also entail a lack of commitment on the part of an individual. However, polar interrogatives differ in that they reflect *ignorance* rather than *doubt*. As mentioned earlier, we have to put aside embedded interrogatives.

¹⁵In [Maho \(2009\)](#), [Hammarström \(2019\)](#) Subi is listed as JD64. There appears to be some confusion about distinct languages called Subi in the region. The Subi discussed here is more likely in within JE20 (according to native speaker linguists), and so I have chosen to give it the code “JD26.”

¹⁶[Hyman & Watters \(1984\)](#) assume that this restriction is “grammatical” (i.e., synchronically idiosyncratic). I respectfully find this to be an unsatisfying solution. Indeed, as Hyman & Watters note, such restrictions tend to occur in “backgrounded” (256) clauses, already suggesting a link to information structure. Moreover, the fact that this same restriction keeps appearing in unrelated languages begs a more general explanation.

(22) Subi (JE26) (Cyprian Vumilia, p.c.)

- a. bha-ka-tu-gambira nkikwo tu-liku-bha-tel-era endulu
2SM-PST-1PL.OM-tell COMP 1PL.SM-PRES-2OM-make-APPL noise
'They told us that we are making noise to them.'
- b. bha-ka-tu-gambira ngu ni-tu-bha-tel-era endulu
2SM-PST-1PL.OM-tell COMP FOC-1PL.SM-2OM-make-APPL noise
'They told us that we are making noise to them.'

An identical pattern is found in languages with a *conjoint/disjoint* system. While it remains debated about the function of the conjoint/disjoint alternation (it is likely to have multiple functions across languages), it is reasonably thought to be connected with predicate (internal) focus: disjoint forms are found when the predicate is in focus, while conjoint forms are found elsewhere (van der Wal 2016).

In languages with a conjoint/disjoint system, disjoint forms of the verb are often banned in certain embedding contexts. It is common to find restrictions inside of nonselected clauses (e.g., relatives and temporal clauses), but we also find that certain selected embedded clauses restrict the verbal morphology. For instance, in Rwanda (JD61) and Rundi (JD62), under the complementizer *kó*, only conjoint forms are found (Ngoboka & Zeller 2016, Nshemezimana & Bostoen 2016). The disjoint form (in this context, marked with *ra-*) is available under *ngo*-clauses.¹⁷

- (23) a. Rwanda (JD61) (Ngoboka & Zeller 2016: 366)
a-ra-vúg-ye ngo a-ra-som-a
1SM-DJ-say-PFV that 1SM-DJ-read-FV
'He says that he reads.'
- b. Rwanda (JD61) (Ngoboka & Zeller 2016: 365)
a-vúg-ye kó a-som-a
1SM-say-PFV that 1SM-read-FV
'He says that he reads.'

The prohibition on predicate focus in an embedded clause must stem from an interaction with constraints on information structure imposed by the clause itself – parallel to the fact that negation, a focus-sensitive operator, also blocks disjoint forms.

¹⁷Note that the main-clause verb is also affected, but the correlations between complementizer and main-clause morphology are less systematic, so I exclude them from the discussion here.

Interestingly, the same dichotomy described for commitment above holds here. Only selected embedded clauses headed by *be*-complementizers or demonstratives block focus operators in the embedded clause. Clauses headed by *say*-complementizers, manner deictics, and pronouns generally have no effect on the information structure – specifically predicate focus – in the embedded clause.¹⁸ Again, I emphasize that not all *be*- or demonstrative complementizers prohibit predicate focus in selected embedded clause. But when it is blocked, the complementizer is always a *be*- or demonstrative complementizer.

3.3 Agreement

The final distinction I mention here is morpho-syntactic. It is well-known that a relatively large number of Bantu languages have so-called *agreeing complementizers*. These are complementizers which exhibit morphological concordance with an element in the matrix clause, almost always the embedding subject,¹⁹ like Bukusu's (JE31c) well-discussed *-li*.

- (24) Bukusu (JE31c) (Diercks 2013: 358)
 ba-ba-ndu ba-bol-el-a Alfredi ba-li a-kha-khil-e
 2-2-people 2s-said-AP-FV 1Alfred 2-that 1s-FUT-conquer
 'The people told Alfred that he will win.'

There is no single lexical source for agreeing complementizers across Bantu languages; they are derived from disparate sources. While the majority of the cases in our sample are related to **-ti*, there are numerous exceptions. Most prominently, there are agreeing complementizers derived from pronouns, discussed in Section 2.3.3.

While productive agreeing complementizers are robustly attested in Bantu, we also find that the agreement has fossilized on many Bantu complementizers. Some languages, like Nyengo (K16), have a complementizer inflected with what is presumably invariant 1sg morphology, (25). (*Nji* is also found throughout the K30 language group.)²⁰

¹⁸ der Wal (2014) makes a similar claim with respect to *situative* clauses; she is largely concerned with nonselected adverbial clauses.

¹⁹ Gluckman (2023) reports that certain speakers of Nyala East (JE32f) permit an implicit subject to control the agreement of the complementizer *-chi*, though he notes that this is the exception, not the rule.

²⁰ Indeed, this appears to be an innovation in some dialects of Rwanda/Rundi (JD61/62). In the "standard" languages, *-ti* is fully agreeing. Based on conversations with speakers, however, some dialects only use *nti*, which can only plausibly be a fossilized 1st-person form.

- (25) Nyengo (K16) (Lisimba 1982: 166)
 tu-faa nji tu-ful-e
 1PL.SM-want COMP 2PL.SM-beat-SBJV
 ‘We want to beat.’

In other languages, what is fossilized appears to be 3rd person agreement. In Lamba (M54) in (26a), *ati* likely bears 3rd singular (Class 1) agreement. *Baati* in Nyakyusa (M31a) in (26b) shows 3rd person plural fossilized agreement.

- (26) a. Lamba (M54) (Doke 1922: 144)
 na li bwene ati ta wa pyangile
 I PST see COMP NEG they sweep
 ‘I saw that they did not sweep.’
 b. Nyakyusa (M31a) (Persohn 2017: 315)
 ... “keet-a ɔ-t-ile baati n-heesya gw-itu!? ...”
 look-FV 2SG-say-PFV hearsay 1-guest 1-POSS.1PL
 ‘...“Look, you said he is our guest!?”’

Pronominal complementizers, too, show similar variation. The invariant complementizers *mbo* and *ngo* discussed above are also complementizers with fossilized agreement. *Mbo* has 3rd plural (Class 2) invariant “agreement” while *ngo* is reconstructed in Botne (1995) as a complementizer/evidential particle with invariant 3rd singular (Class 1) morphology.²¹

In our sample, systemically excluded from the class of agreeing complementizers – including those with fossilized agreement – are *be*-complementizer and demonstrative complementizers.²² For demonstratives, while these do reflect noun classes, they (i) never *change* noun class within a language, (ii) never reflect 1st, 2nd, or 3rd person (human) morphology, and (iii) never reflect agreement with something overt in the superordinate clause. More strikingly, the lack of person/number marking is particularly relevant for *be*-complementizers, which can and often do, exhibit agreement synchronically.

²¹Nduumo (B63) and Mbete (B61) have a more unique situation, as reported in Adam (1954). With 3rd person (singular/plural) subjects, invariant *ndi* is the complementizer used. With 1st and 2nd person (singular/plural) subjects, *mbidi* (for Nduumo) and *meti* (for Mbete) are used. Thus, these languages show partially fossilized agreement, collapsing the speech-act participants.

²²Similatives also do not show evidence of co-variation with the main-clause subject in our language sample.

3.4 The (lack of) selection

Most typological studies of complementizers start by looking at the interaction between complementizer/complementation choice and the lexical semantics of the embedding predicate. Such studies are interested in whether there are selectional restrictions on complementation. Certainly, selection plays an important role in Bantu languages: we find a general divide between those predicates which embed nonfinite clauses (e.g., modals, aspectuals), and those which embed finite clauses (belief verbs, utterances predicates, etc.). Within the finite clauses, we also find that there are general divisions between interrogative and non-interrogative clauses. But in general, there is limited evidence that the lexical class of the embedding predicate plays a direct role in the choice of the complementizers described in this paper. Works that note correlations between complementizer choice and category of the embedding predicate are relatively sparse (cf. Givón & Kimenyi 1974; Van de Velde 2008; Finholt & Gluckman 2023).

I believe, in fact, that the importance of selection is often over-emphasized, or at the very least, not the main factor in complementizer selection. The issue is one of confirmation bias. The starting place for all investigations of Bantu complementizers is always lexical class of the embedding verb – a sensible place to start (see for instance Safir 2008). But this sometimes leads to the conclusion that any distinctions in complementizers that arise are solely attributable to the lexical class of the embedding predicate. But more often than not in the descriptive sources (even beyond Bantu languages) and in my own fieldwork, the same embedding predicate may appear with multiple kinds of embedded clause, yielding slightly distinct readings. Thus, Myers (1975: 189) reports that in Chewa (N31b) “the choice of complementizers is controlled in part by the syntactic elements of the sentence, and in part by the pragmatics of its use.”

In other languages, there is essentially no effect of predicate class. In Kamba (E55) the two complementizers *kana* and *atĩ* are in free variation (Myers 1975: 187). Likewise, in Nyala East (JE32f), the complementizers *mbo*, *-chi*, and *vachi* are entirely interchangeable under all embedding verbs (and nouns).²³

²³It is worth noting that Mood of the embedded clause only has an indirect effect on the choice of complementizer as well. That is, while many complementizers correlate with subjunctive mood, the predicting factor for mood choice is almost always lexical class of the embedded verb. For instance, desire predicates like *want* strongly prefer subjunctive mood (particularly with disjoint subjects), regardless of the complementizer used to introduce the embedded clause. (See e.g., discussion concerning Gyeli (A801) in Grimm 2021.) Insomuch as the embedding verb influences the choice of complementizer, then it may appear that complementizer and mood are interacting.

Finally, there are additional factors that can influence complementizer choice, though these are restricted in their distribution. [Letsholo & Safir \(2019\)](#) discuss instances of Voice affecting the choice of complementizer in Nande (JD42) and Kalanga (S16). Similarly, Shona (S10) has both an active and passive complementizer, *kuti* and *kunzi*, respectively ([Dembetembe 1976](#)). In Iwoyo (H16d), as discussed in [Mingas \(1994\)](#), the *say*-complementizer *tí* appears in affirmative statements, but the complementizer *kàní* appears when the main verb is negated. In Fuliiru (JD63), the pronominal complementizer *mbu* is used in “frustrative” constructions ([Van Otterloo 2011](#)). And in Rufumbira (JD62a), as reported in [Nassenstein \(2016\)](#), the choice of *ngo* versus *mbu* is intermixed with language attitudes and identity. These additional distinctions deserve a more detailed crosslinguistic study than I can provide here.

4 Classifying the dichotomy

I summarize my initial findings in Table 2.

Table 2: Summary of the correlations between form and function

	<i>say</i> -comps	manner	pronouns	<i>be</i> -comps	demonstratives
Speaker/subject commitment	weak	weak	weak	strong	strong
Can restrict FOC. in emb. clause	✗	✗	✗	✓	✓
Can show agreement	✓	✓	✓	✗	✗

If these generalizations hold up, it suggests a striking correlation between form and function that deserves an explanation. I suggest the following tentative, intuitive explanation. We start with the assumption that embedded clauses can be related to main clause to various ways. I refer to this as *anchoring*. One way to anchor the embedded clause to the main clause is via an individual: the embedded clause reflects information conveyed by, or held to be true/false by, or rejected by (etc.), some individual, possibly the subject of the embedding verb, or possibly the speaker. For instance, quotation can be anchored to an individual: the speaker of the quoted utterance. This idea can extend to attitudinal embedding as well.

The sentence *Maina believes that it's raining* is judged true if the proposition “It's raining” is among Maina's beliefs.

On the other hand, embedded clauses can be anchored to a situation (or even-tuality). For example, a quotation can be anchored to “the event denoted by the quote” (Güldemann 2008: 362). Again this idea can extend beyond quotation to cover propositions which are judged true or false at a particular time and place. This is the approach taken across a number of theoretical works, including (Aboh 2004, Kratzer 2006, Moulton 2008, Hacquard 2009, Haegeman & Ürögdi 2010, Moulton 2013, Hegarty 2016). The basic idea across these works is that embedded clauses (can) describe propositional content that is evaluated with respect to an event(uality). Conceptually, according to the latter authors' view, in *Maina believes that it's raining*, the embedded proposition “It is raining” is evaluated relative to the beliefs held at the event described by *believe*. The attitude holder *Maina* only plays an indirect role in that it happens to be Maina's beliefs in this case.

Based on this dichotomy, I suggest that the division among Bantu complementizers reflects this individual/situation divide: some complementizers are “perspectival,” anchoring the embedded clause to an individual. *Say*-complementizers, manner deictics, and pronouns all fall into this category: they are all linked morphologically or semantically to an individual's perspective. This can be seen directly in the morphology, through agreement with a local subject. The effect of anchoring to an individual can have semantic effects as well. The weak commitment readings observed in Section 3.1 may be understood as the result a “subjective” attitudinal stance. That is, using a perspectival complementizer signals that the embedded clause is evaluated relative to a particular individual who is not the speaker, and is not common/shared knowledge.

The other complementizers, *be*-complementizers and demonstratives, are “situational,” anchoring the embedded clause to a situation. The lack of agreement follows naturally. The interaction with predicate focus also follows on the assumption that if the complementizer introduces or foregrounds a salient situation, that situation can interact with a foregrounded (focused) situation in the embedded clause. Indeed, clear evidence of the relationship between predicate focus and complementizers comes from shared morphology: the predicate focus operator in some languages *ni* is also a complementizer, as discussed in Section 2.2.

To account for stronger speaker commitment, we can make reference to *definite* situations, which commit the speaker to the belief in a situation that is relevant in evaluating the embedded clause. As discussed in Özyıldız (2017), Bochnak & Hanink (2021) (for non-Bantu languages), presupposing a situation in which

the embedded clause is true can lead to a factive inference, i.e., a “stronger” commitment towards the embedded proposition.

5 Conclusion

In this paper, I have provided a preliminary typology of Bantu complementizers. Even with limited data, this study reveals rich variation within the broader family. Additionally, I have explored three factors that correlate with complementizer choice in languages which have more than one complementizer. Based on the available data, I suggested a dichotomy between perspectival and situational complementizers. Though I caution again that these results need further evaluation, with closer inspection of Bantu complementizer systems, I nonetheless believe that the emerging patterns offer an important and novel window into the typology and function of complementizers crosslinguistically.

Abbreviations

1–20	noun classes/person	INF	infinitive
AGR	agreement	IPFV	imperfective
AM	assertive marker	ITIVE	itive
APPL/AP	applicative	LNK	linker
ASP	aspect	LOC	locative
AUG	augment	LT	linking tone
AUX	auxiliary	NEAR.FUT	near future
CL/NC/NP	noun class marker	NEG	negative
COMP/CMP	complementizer	NUCL	nucleative
COMPL	completive	OC/OM/O	object
CONS	consecutive		marker/concord
COP	copula	P	person marker
DEM	demonstrative	PFV/PRF	perfective
DJ	disjoint	PL	plural
DUB	dubitative	POSS	possessive
FPAST	far past	PP	pre-prefix or
FUT	future		pronominal prefix
FV	final vowel	PR	pronoun
HAB	habitual	PRES	present
IDEO	ideophone	PRT	particle
IMP	imperative	PST	past

QUOT/QV	quotative	SB/SBV/SBJV	subjunctive
REF	referential	SG	singular
RPAST/RCPAST	recent past	TAM	tense/aspect/mood
s/SA/SC/SM	subject agreement/ concord/marker		

Acknowledgements

Thanks to Aron Finholt and other members of KUBantu at the University of Kansas for helping to collect, organize, and understand the data. Thanks especially to Jacques Alimasi, Antoinette Bifuko, Fudjo Jeannot, Raphael Kayembe, JP Sabimpa, and Cyprian Vumilia for their time sharing their languages with me. And thanks as well to audience members at ACAL 54 for their helpful feedback. This work was supported by a grant from the National Science Foundation (2140837).

References

- Aboh, Enoch. 2004. *The morphosyntax of complement-head sequences: Clause structure and word order patterns in Kwa*. Oxford: Oxford University Press.
- Adam, Jérôme J. 1954. *Grammaire composée mbede, ndumu, duma*. Montpellier: Imprimerie Charité.
- Bastin, Yvonne, André Coupez, Evariste Mumba & Thila C. Schadeburg (eds.). 2002. *Bantu lexical reconstructions 3 / reconstructions lexicales bantoues 3*. Tervuren: Royal Museum for Central Africa.
- Bochnak, Ryan & Emily Hanink. 2021. *Clausal embedding in Washo: Complementation vs. modification*. manuscript.
- Botne, Robert. 1995. The pronominal origin of an evidential. *Diachronica* 12(2). 201–221.
- Botne, Robert. 2003. Lega (Beya dialect) (D25). In Mark Van de Velde, Koen Bostoen, Derek Nurse & Gérard Philippson (eds.), *The Bantu Languages*, 2nd edn., chap. 22, 422–445. London: Routledge.
- Botne, Robert. 2020. Evidentiality in African languages. In Chungmin Lee & Jinho Park (eds.), *Between evidentials and modals*, 460–501. Leiden: Brill.
- Chappell, Hilary. 2008. Variation in the grammaticalization of complementizers from verba dicendi in Sinitic languages. *Linguistic Typology* 12(1). 1–50.
- Cristofaro, Sonia. 2003. *Subordination*. Oxford: Oxford University Press.
- De Rop, Albert. 1958. *Grammaire du lomongo (phonologie et morphologie)*. Léopoldville: Éditions de l'Université.

- Dembetembe, Norris Clemens. 1976. *The syntax of sentential complements in Shona*. University of London. (Doctoral dissertation).
- der Wal, Jenneke Van. 2014. Subordinate clauses and exclusive focus in Makhuwa. In Rik van Gijn, Jeremy Hammond, Dejan Matić, Saskia van Putten & Ana Vliacy Galucia (eds.), *Information structure and reference tracking in complex sentences*, 45–69. Amsterdam: John Benjamins.
- Devos, Maud & Koen Bostoen. 2012. Bantu do/say polysemy and origins of the quotative in Shangaci. *African Linguistica* 18. 97–132.
- Diercks, Michael. 2013. Indirect agree in Lubukusu complementizer agreement. *Natural Language & Linguistic Theory* 31(2). 357–407.
- Diessel, Holger & Merlijn Breunese. 2020. A typology of demonstrative clause linkers. In Åshild Næss, Anna Margetts & Yvonne Treis (eds.), *Demonstratives in discourse*, 305–341. Berlin: Language Science Press.
- Djäv, Kajsa. 2019. *Factive and assertive attitude reports*. Philadelphia: University of Pennsylvania. (Doctoral dissertation).
- Doke, Clement. 1922. *The grammar of the Lamba language*. London: K. Paul, Trench, Trubner & Company.
- du Plessis, Jacobus A. 1989. Distribution of the complementizer *ukuba* in the Xhosa sentence. *South African Journal of African Languages* 9(2). 43–51.
- Edelsten, Peter, Hannah Gibson, Rozenn Guérois, Gastor Mapunda, Lutz Marten & Julius Taji. 2022. Morphosyntactic variation in Bantu: Focus on East Africa. *Journal of the Language Association of Eastern Africa* 1(1). <https://scholarship.claremont.edu/jlaea/vol1/iss1/4>.
- Englebretson, Robert (ed.). 2015. *A basic sketch grammar of Gikūyū* (Rice Working Papers in Linguistics 6). Houston.
- Finholt, Aron. 2024. *Non-verbal predication in Eastern Bantu*. PhD, University of Kansas.
- Finholt, Aron & John Gluckman. 2023. A corpus analysis of Swahili's dual-complementizer system. *Journal of African Languages and Linguistics* 44(1). 25–48.
- Gibson, Hannah, Rozenn Guérois & Lutz Marten. 2018. Variation in Bantu copular constructions. In María J. Arché, Antonio Fábregas & Rafael Marín. (eds.), *The grammar of copulas across languages*, 213–242. Oxford: Oxford University Press.
- Gilliard, Louis Octave. 1928. *Grammaire pratique loutomba*. Bruxelles: Editions de l'Essorial.
- Givón, Talmy & Alexandre Kimenyi. 1974. Truth, belief and doubt in Kinyarwanda. *The Papers from the Fifth Annual Conference on African Linguistics* 5. 95–114.

- Gluckman, John. 2021. Null expletives and embedded clauses in Logoori. *Syntax* 24. 334–375.
- Gluckman, John. 2023. Perspectival domains in nouns and clauses. *Natural Language & Linguistic Theory* 41. 159–223.
- Grimm, Nadine. 2021. *A grammar of Gyeli*. Berlin: Language Science Press.
- Güldemann, Tom. 2002. When ‘say’ is not say. In Tom Güldemann & Manfred von Roncador (eds.), *Reported discourse: A meeting ground for different linguistic domains*, 253–288. Amsterdam: John Benjamins.
- Güldemann, Tom. 2003. Present progressive vis-à-vis predication focus in Bantu – A verbal category between semantics and pragmatics. *Studies in Language* 27(2). 323–360.
- Güldemann, Tom. 2008. *Quotative indexes in African languages: A synchronic and diachronic survey* (Empirical Approaches to Language Typology 34). Berlin: Mouton de Gruyter.
- Güldemann, Tom. 2013. *The relation between predicate operator focus and theticity in Bantu*. Handout from the Ghent-Berlin Workshop *Information structure in Bantu*. Humboldt University.
- Gunnink, Hilde. 2018. *A grammar of Fwe: A Bantu language of Zambia and Namibia*. University of Ghent. (Doctoral dissertation).
- Guthrie, Malcolm. 1970a. *Comparative Bantu*, vol. 3. England: Gregg International Publishers.
- Guthrie, Malcolm. 1970b. *Comparative Bantu*, vol. 4. England: Gregg International Publishers.
- Hacquard, Valentine. 2009. On the interaction of aspect and modal auxiliaries. *Linguistics and Philosophy* 32(3). 279–315.
- Haegeman, Liliane & Barbara Ürögi. 2010. Referential CPs and DPs: An operator movement account. *Theoretical Linguistics* 36(2/3). 111–152.
- Halpert, Claire. 2016. *Argument licensing and agreement*. Oxford: Oxford University Press.
- Halpert, Claire. 2019. How to be an embedded clause: say complementizers in Bantu. In Ryan Bochnak, Miriam Butt, Erlinde Meertens & Mark-Matthias Zymla (eds.), *Proceedings of TripleA 5: Fieldwork perspectives on the semantics of African, Asian and Austronesian languages*, 31–46. Universitätsbibliothek Tübingen, Publikationssystem.
- Hammarström, Harald. 2019. An Inventory of Bantu languages. In Mark Van de Velde, Koen Bostoen, Derek Nurse & Gérard Philippson (eds.), *The Bantu languages, second edition*, 17–79. New York: Routledge.
- Hegarty, Michael. 2016. *Modality and propositional attitudes*. Cambridge: Cambridge University Press.

- Hopper, Paul & Elisabeth Closs Traugott. 1993. *Grammaticalization*. Cambridge: Cambridge University Press.
- Hyman, Larry & John Watters. 1984. Auxiliary focus. *Studies in African Linguistics* 15(3). 233–273.
- Johnson, Amandus. 1931. *Mbundu English-Portuguese dictionary: With grammar and syntax*. Philadelphia: International Printing Co.
- Kaburo, Diana K. 2022. A syntactic analysis of raising structures in Kĩmũthambĩ. *Journal of Linguistics, Literary, and Communication Studies* 1(2). 44–53.
- Kawasha, Boniface. 2006. The structure of complement clauses in Lunda. *Studies in African Linguistics* 35(1). 1–32.
- Kawasha, Boniface. 2007. Subject-agreeing complementizers and their functions in Chokwe, Luchazi, Lunda, and Luvale. In Doris Payne & Jaime Peña (eds.), *Selected proceedings of the 37th Annual Conference on African Linguistics*, 180–190.
- Kihara, Patrick. 2017. A functional analysis of atĩ and its variants atĩrĩrĩ and atĩrĩ as complementisers and discourse markers in Gĩkũyũ. *The University of Nairobi Journal of Language and Linguistics* 6. 102–127.
- Kratzer, Angelika. 2006. *Decomposing attitude verbs*. accessed Accessed on Jan 25, 2018 at <http://semanticsarchive.net/Archive/DcwY2JkM/attitude-verbs2006.pdf>.
- Kuteva, Tania, Bernd Heine, Bo Hong, Haiping Long, Heiko Narrog & Seongha Rhee. 2019. *World lexicon of grammaticalization, second edition*. Cambridge: Cambridge University Press.
- Lax, Beverle Michael. 1996. *The verbal system of Kigiryama*. University of Wisconsin-Madison. (Doctoral dissertation).
- Letsholo, Rose & Ken Safir. 2019. Complement clause C-agreement beyond subject phi-agreement in Ikalanga. In Samson Lotven, Silvina Bongiovanni, Phillip Weirich, Robert Botne & Samuel Gyasi Obeng (eds.), *African linguistics across the disciplines* (Contemporary African Linguistics 5), 159–186. Berlin: Language Science Press.
- LIDEMO. 2010. *Mwani: Grammatical sketch*. SIL Moçambique.
- Lisimba, Mukumbuta. 1982. *Luyana dialectology*. University of Wisconsin-Madison. (Doctoral dissertation).
- Maho, Jouni Filip. 2009. *NUGL Online: The online version of the New Updated Guthrie List, a referential classification of the Bantu languages*. Online at https://brill.com/fileasset/downloads_products/35125_Bantu-New-updated-Guthrie-List.pdf.
- Masatu, Amanda Swift. 2015. *Suba-Simbiti narrative discourse*. Dallas: SIL International.

- Meeussen, Achille Emile. 1967. Bantu grammatical reconstructions. *Africana Linguistica* 3. 79–121.
- Mingas, Amélia Arlete. 1994. *Etude grammaticale de l'iwoyo*. CNRS. (Doctoral dissertation).
- Morrison, Michelle Elizabeth. 2011. *A reference grammar of Bena*. Rice University. (Doctoral dissertation).
- Moulton, Keir. 2008. Clausal complementation and the *wager*-class. In Anisa Schardl, Martin Walkow & Muhammad Abdurrahman (eds.), *Proceedings of the Northeast Linguistics Society (NELS) 38*. Cambridge: Cascadia Press.
- Moulton, Keir. 2013. Not moving clauses: Connectivity in clausal arguments. *Syntax* 16(3). 250–291. DOI: [10.1111/synt.12007](https://doi.org/10.1111/synt.12007).
- Mous, Matern. 2004. *A grammatical sketch of Mbugwe: Bantu F34, Tanzania* (Grammatical Analyses of African Languages). Cologne: Rüdiger Köppe Verlag.
- Myers, Amy. 1975. Complementizer choice in selected Eastern Bantu languages. In *Proceedings of the Sixth Conferences on African Linguistics*, 184–193. OSU Working Papers in Linguistics 20.
- Nassenstein, Nico. 2016. *Speaking with a difference: Border thinking in Rufumbira*. Universität zu Köln. (Doctoral dissertation).
- Ngoboka, Jean Paul & Jochen Zeller. 2016. The conjoint/disjoint alternation in Kinyarwanda. In Jenneke Van der Wal & Larry M. Hyman (eds.), *The conjoint/disjoint alternation in Bantu*, 350–389. Berlin: De Gruyter/Moulton.
- Nicolle, Steve. 2014. *Digo narrative discourse* (SIL Language and Culture Documentation and Description 26).
- Nicolle, Steve, Brittney Balfour, Bryanne Friesen, Nick Toews & Jesse Workman. 2018. *Selected narrative discourse features in Kisi, a Bantu language of Tanzania* (CanIL Electronic Working Papers 4).
- Nshemezimana, Ernest & Koen Bostoen. 2016. The conjoint/disjoint alternation in Kirundi (JD62): A case for its abolition. In Jenneke Van der Wal & Larry Hyman (eds.), *The conjoint/disjoint alternation in Bantu*, 390–425. Berlin: De Gruyter Mouton.
- Nurse, Derek. 2008. *Tense and aspect in Bantu*. Oxford: Oxford University Press.
- Özyıldız, Deniz. 2017. Attitude reports with and without true belief. In *Proceedings of SALT 27*, 397–417.
- Persohn, Bastian. 2017. *The verb in Nyakusa: A focus on tense, aspect and modality* (Contemporary African Linguistics 2). Berlin: Language Science Press.
- Pietraszko, Asia. 2019. Obligatory CP nominalization in Ndebele. *Syntax* 22(1). 66–111.

- Roth, Tim. 2018. *Aspect in Ikoma and Ngoreme: A comparison and analysis of two Western Serengeti Bantu languages*. University of Helsinki. (Doctoral dissertation).
- Rugege, Geoffrey. 1984. *A study of Kinyarwanda complementation*. University of Illinois Urbana-Champaign. (Doctoral dissertation).
- Rugemalira, Josephat. 2005. *A grammar of Runyambo*. University of Dar es Salaam: Languages of Tanzania Project.
- Sabimana, Firmard. 1986. *The relational structure of the Kirundi verb*. Indiana University. (Doctoral dissertation).
- Safir, Ken. 2008. *Afranaph Database*. [https : / / www . afranaphproject . afranaphdatabase.com/](https://www.afranaphproject.afranaphdatabase.com/).
- Sørensen, Marie-Louise Lind & Alena Witzlack-Makarevich. 2020. Clausal complementation in Ruuli (Bantu, JE103). *Studies in African Linguistics* 49(1). 84–110.
- Van de Velde, Mark. 2008. *A grammar of Eton*. Berlin: Mouton de Gruyter.
- van der Wal, Jenneke. 2016. What is the conjoin/disjoint alternation? Parameters of crosslinguistic variation. In Jenneke Van der Wal & Larry M. Hyman (eds.), *The conjoin/disjoint alternation in Bantu*, 14–60. Berlin: De Gruyter.
- Van Otterloo, Roger. 2011. *The Kifuliiru language: A descriptive grammar*, vol. 2. Dallas: SIL International.